

STACK (Method Area)

main(String[] args)

Stack Frame

Variable	Value
args	reference to String array
calc	0x2001

calc is a reference variable stored in STACK. It holds the address of the Calculator object in HEAP.

Legend

0x2001 = Memory Address
(Example)

HEAP (Objects Area)

Object: Calculator (created by new Calculator())

Address: 0x2001

Field (Attribute)	Value
num1	0.0
num2	0.0
result	0.0

Object States During Execution

1. After new Calculator()

Field	Value
num1	0.0
num2	0.0
result	0.0

2. After setNumbers(20, 5)

Field	Value
num1	20.0
num2	5.0
result	0.0

3. After add()

Field	Value
num1	20.0
num2	5.0
result	25.0

4. After subtract()

Field	Value
num1	20.0
num2	5.0
result	15.0

5. After multiply()

Field	Value
num1	20.0
num2	5.0
result	100.0

6. After divide()

Field	Value
num1	20.0
num2	5.0
result	4.0

7. After setNumbers(10, 0)

Field	Value
num1	10.0
num2	0.0
result	4.0 (result unchanged)

8. After divide()

Field	Value
num1	10.0
num2	0.0
result	4.0 (Cannot divide by zero)

Key Points

- New object (Calculator) is created in HEAP.
- Reference variable (calc) is stored in STACK.
- All attributes (fields) of the object live in HEAP.
- Methods operate on the object in HEAP through the reference.
- Any change in object state affects the same object in HEAP.